Effects of acute administration of caffeine on vascular function.

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Caffeine is the most widely used pharmacologic substance in the world. It is found in common nonessential grocery items (e.g., coffee, tea, cocoa, and chocolate). The effects of caffeine on cardiovascular diseases, including hypertension, remain controversial, and there is little information on its direct effect on vascular function. The purpose of this study was to determine the effect of caffeine on endothelial function in humans. This study was a double-blind, randomized placebo and active drug study. Forearm blood flow (FBF) responses to acetylcholine (ACh), an endothelium-dependent vasodilator, and to sodium nitroprusside, an endothelium-independent vasodilator, were evaluated in healthy young men before and after the oral administration of caffeine 300 mg ($n = 10$) or placebo ($n = 10$). FBF was measured by using a strain-gauge plethysmograph. Caffeine significantly increased systolic and diastolic blood pressures by 6.0 +/- 6.0 and 2.6 +/- 3.1 mm Hg ($p < 0.05$), respectively, but did not alter heart rate or baseline FBF. Caffeine augmented the FBF responses to ACh from 21.2 +/- 7.1 to 26.6 +/- 8.1 ml/min/100 ml tissue ($p < 0.05$), whereas sodium nitroprusside-stimulated vasodilation was not altered by caffeine administration. The intra-arterial infusion of N(G)-monomethyl-L-arginine, a nitric oxide synthase inhibitor, abolished the caffeine-induced augmentation of FBF response to ACh. In the placebo group, the ACh- and sodium nitroprusside-stimulated vasodilation was similar before and after the follow-up period. In conclusion, these findings suggest that the acute administration of caffeine augments endothelium-dependent vasodilation in healthy young men through an increase in nitric oxide production.